SecureDT - AI Model Guide

Training, explainability and offline fallback

The risk model predicts the security level of an incoming scheduled task from labelled synthetic workloads.

Training

```
python scripts/train_ai.py
```

Random Forest via RandomizedSearchCV on ai_training_10000.csv; writes models/risk_model.pkl and models/risk_metrics.json (ROC-AUC, F1, importances).

Runtime behaviour

- Predicts level (safe/trained/critical) + confidence
- Shap feature reasons when shap is installed
- Deterministic rule fallback when the model file is missing
- JSON-serialisable metrics via /api/ai/analyze and export

Validation

tests/test_engine.py verifies schema and model-or-fallback; system-health reports model readiness.